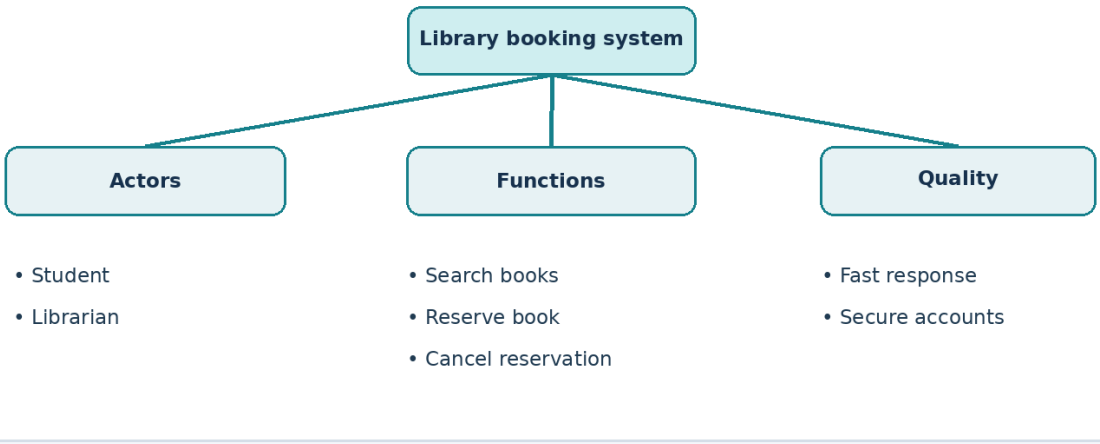
SOFTWARE PROJECT MANAGEMENT

Lab 01 · Visual Requirements Engineering with draw.io

A 90-minute individual practical for first-time diagramming students.

COURSE	**Software Project Management**
LAB FORMAT	Individual work only
PLATFORM	draw.io in a web browser
DURATION	90 minutes
INSTRUCTOR	Dr. Shabir Hussain
INSTITUTION	Zhongzhou University, China

EXAMPLE — REQUIREMENTS MIND MAP



What students will produce

Three original, editable diagrams: (1) requirements mind map, (2) UML use-case diagram, and (3) simple process flowchart. The semester team project is assigned separately on the final page.

Class preparation: a computer, internet connection, browser, and a local folder for saving files.

01 | Lab purpose and learning outcomes

This lab introduces a practical way to turn a software idea into clear visual requirements. Before writing code, software engineers need to understand who will use a system, what the system must do, and what steps occur in an important process.

By the end of this lab you should be able to

☐ Open the online draw.io editor and create a new editable file.

☐ Recognize the canvas, shape library, connectors, menus, and formatting tools.

☐ Separate users (actors), functional requirements, and quality requirements.

☐ Create a simple UML use-case diagram and a flowchart.

☐ Save an editable .drawio version and export a readable PNG image.

90-minute laboratory schedule

TIME	ACTIVITY	OUTPUT
00–15 min	Open draw.io; practice shapes and saving	Practice file
15–35 min	Task 1 — requirements mind map	01 Requirements
35–60 min	Task 2 — UML use-case diagram	02 UseCase
60–80 min	Task 3 — booking process flowchart	03 Flowchart
80–90 min	Export, check, and submit individual work	Six files total

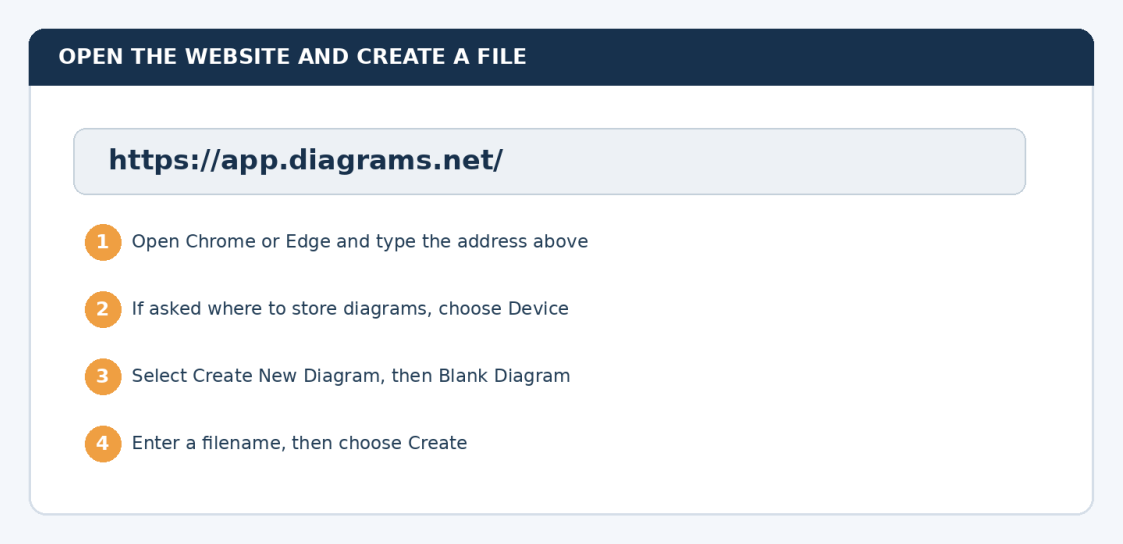
Important rules

All three exercises are INDIVIDUAL. Do not submit a teammate's diagram. You may ask the instructor questions, but you must create and save your own three diagrams. The final semester project activity is separate and is not included in the individual lab mark.

Software used

Online editor: https://app.diagrams.net/ (also known as draw.io). No account, programming tool, or desktop application is required for this lab. When prompted to choose storage, select Device to save on your computer.

02 | First-time setup: Step-by-step



01 Open the browser
Click the Chrome or Microsoft Edge icon on the computer. Click the address bar at the top of the window.

02 Visit the editor
Type https://app.diagrams.net/ and press Enter. Wait for the site to load. If the website is blocked or unavailable, notify the instructor.

03 Choose where to save
If a storage selection screen appears, choose Device. Some browsers may show a privacy or storage notice first; continue to the editor. Do not connect a personal cloud account for this lab.

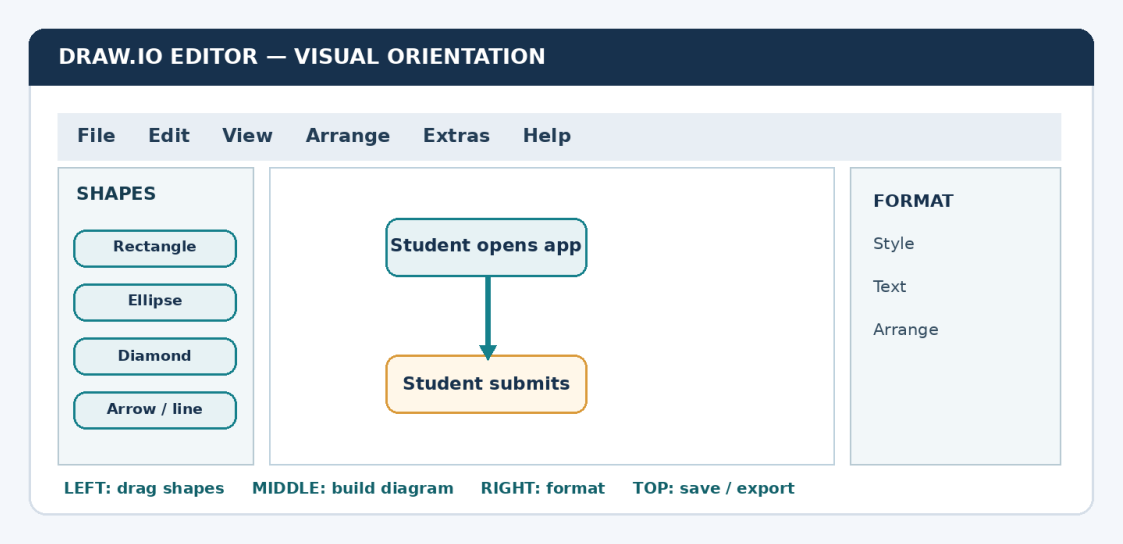
04 Start a blank diagram
Choose Create New Diagram (or File > New). Choose Blank Diagram. Enter Practice_YourStudentID as the filename and click Create.

05 Save the editable diagram
Choose File > Save or press Ctrl+S. If your browser asks where to download the file, choose your lab folder. Confirm the file ends in .drawio. Menu wording can vary slightly by editor version.

Create your lab folder now

In File Explorer, make a folder named SPM_Lab01_YourStudentID. Keep every .drawio file and every PNG image in this folder. Replace YourStudentID with your actual student ID.

03 | Understanding the draw.io workspace



AREA	WHAT IT DOES	HOW YOU USE IT
Top menu	Create, save, export and arrange	Click File for saving and image export
Left shape library	Contains rectangles, circles, arrows and UML shapes	Click and drag a shape onto the canvas
Center canvas	Your drawing area	Move, resize and connect shapes here
Right format panel	Changes colors, text and layout	Select one shape, then adjust its style

Five-minute practice

01 Place a shape

Drag a rectangle from the left panel to the middle of the white canvas.

02 Write inside

Double-click the rectangle and type: Student registers. Click an empty area to finish editing.

03 Connect shapes

Drag a second rectangle below it. Hover over the first shape and drag a connector arrow toward the second shape. Verify that the line attaches to both shapes.

04 Make your work readable

Select a shape, increase its width if the text wraps badly, and use a consistent fill color. Use arrows to show direction.

A diagram is not complete just because it looks attractive: words must be accurate, labels readable, and connectors meaningful.

04 | Individual Task 1 — Requirements mind map

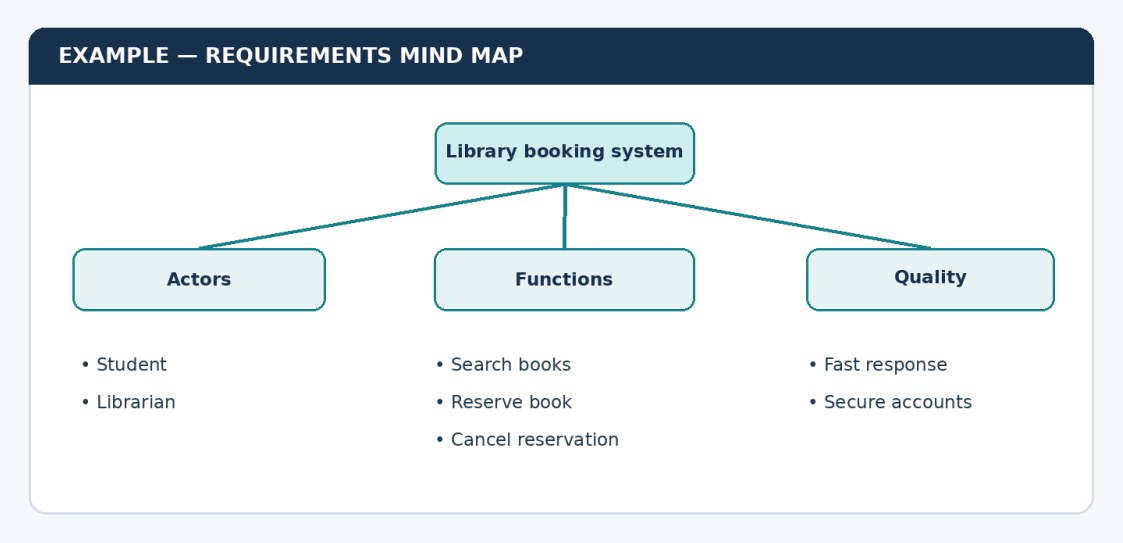
TIME: 20 MINUTES | FILE NAME: 01_Requirements_YourStudentID.drawio

Scenario

A university wants a small Library Booking System. A student should search for a book, reserve an available copy, and cancel a reservation. A librarian should view reservations and update whether books are available. Your job is to organize the initial software requirements before the development team begins coding.

What this diagram is used for

A requirements mind map groups important ideas in one place. It helps a project manager discover missing functions, identify system users, and discuss expectations with stakeholders early.



Build it in draw.io

01 Create a new diagram
Click File > New > Blank Diagram. Name it 01_Requirements_YourStudentID.

02 Make the central topic
Drag a rounded rectangle to the top-center of the canvas. Type Library Booking System.

03 Add three branches
Add three smaller rectangles below it: Actors, Functional Requirements, and Non-functional Requirements. Connect each to the main topic.

04 Add required details
Under Actors: Student, Librarian. Under Functional Requirements: Search books, Reserve book, Cancel reservation, View reservations, Update availability. Under Non-functional Requirements: Easy to use, Secure login, Fast response.

05 Review and save
Check that all text is visible and each item belongs to the correct branch. Save the editable file and export a PNG using File > Export as > PNG.

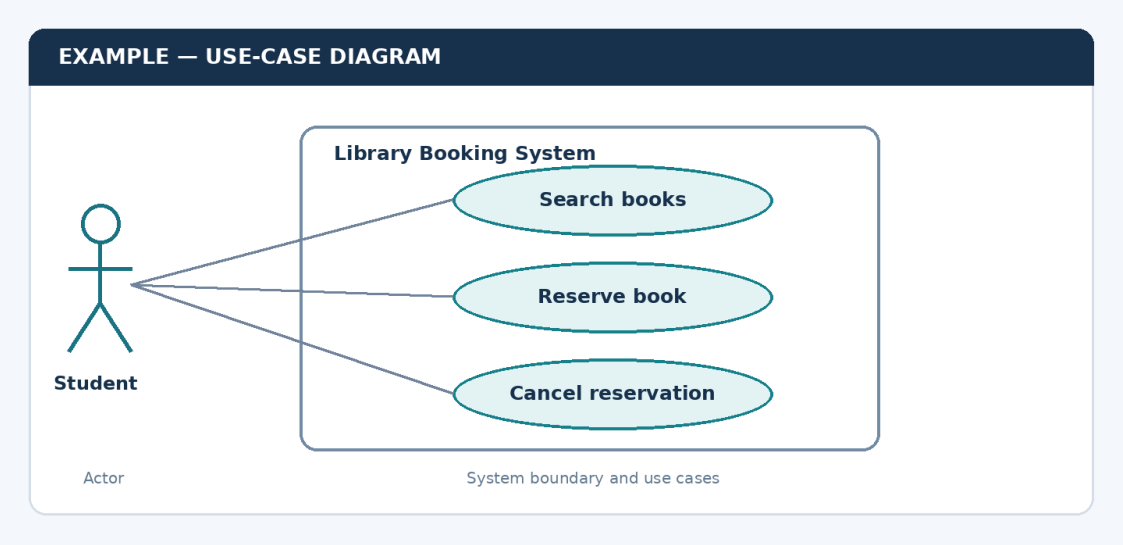
Self-check: Can you identify at least two users, five system functions, and three quality expectations? These are different kinds of requirements.

05 | Individual Task 2 — UML use-case diagram

TIME: 25 MINUTES | FILE NAME: 02_UseCase_YourStudentID.drawio

What a use-case diagram shows

A use-case diagram shows WHO interacts with a system and WHAT the system lets them do. An actor is a role, not a specific person. A use case is a service or action provided by the system. The rectangle is the system boundary.



Build it in draw.io

01 Create a new file
File > New > Blank Diagram. Name it 02_UseCase_YourStudentID.

02 Enable UML shapes
In the left shape library, select More Shapes (if needed), enable UML or UML 2.5, and click Apply. If UML shapes do not appear, use a stick figure, ellipses and rectangle from the basic shape library.

03 Draw the system boundary
Place a large rectangle in the middle of the canvas. Label it Library Booking System. Keep use cases inside this rectangle.

04 Add actors outside
Place Student to the left and Librarian to the right. They must remain outside the system boundary.

05 Add six use cases
Inside the rectangle, add ellipses labeled Search books, Reserve book, Cancel reservation, View reservation, Update book availability, and Check reservation status.

06 Connect actors to their actions
Connect Student to Search, Reserve, Cancel, and Check reservation status. Connect Librarian to View reservation and Update book availability. Use straight association lines (arrowheads are not needed).

07 Save and export
Save the .drawio file. Then click File > Export as > PNG > Export, and save the PNG in your lab folder.

Common mistakes to avoid

Do not put the Student actor inside the system rectangle. Do not describe use cases as implementation tasks such as “Write Python code”. Use meaningful verb phrases (“Reserve book”) and ensure every use case is connected to at least one actor.

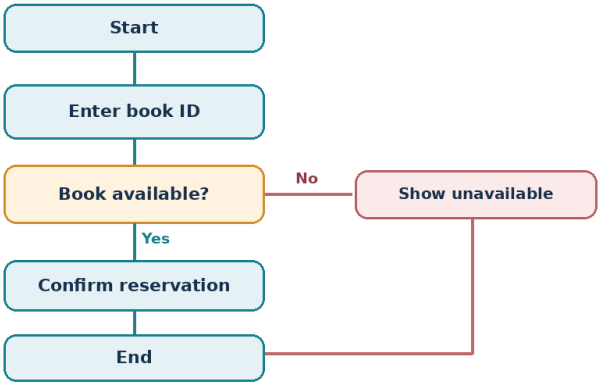
06 | Individual Task 3 — Reservation process flowchart

TIME: 20 MINUTES | FILE NAME: 03_Flowchart_YourStudentID.drawio

What a flowchart is used for

A flowchart describes the ORDER of operations and the decisions in a process. Developers use it to reason about workflow, exceptional cases, and test cases before implementation. A decision must show more than one possible path.

EXAMPLE — PROCESS FLOWCHART



Build it in draw.io

01 Open a blank file
Create a new blank diagram named 03_Flowchart_YourStudentID.

02 Place the main steps vertically
Add Start (oval), Enter book ID (rectangle), Book available? (diamond), Confirm reservation (rectangle), and End (oval).

03 Connect the successful path
Draw arrows Start → Enter book ID → Book available?. From the diamond, label the downward arrow Yes, then connect to Confirm reservation → End.

04 Show the alternative path
Place a rectangle to the right labeled Show unavailable message. Draw an arrow from the decision diamond to it, label the arrow No, and connect that message to End.

05 Check both routes
Follow the diagram with your finger: what happens if the book is available? What happens if it is unavailable? Both paths must reach End.

06 Save and export
Save as .drawio. Export as PNG. Make sure arrows, Yes/No labels and all text are legible.

Mini reflection

In your own words, explain: (a) why a diamond is used for “Book available?”, and (b) what would go wrong if the No branch were missing. Your instructor may ask you to explain this verbally.

07 | Submit and verify your individual lab work

Required files — six per student

EDITABLE SOURCE	IMAGE EXPORT
01_Requirements_YourStudentID.drawio	01_Requirements_YourStudentID.png
02_UseCase_YourStudentID.drawio	02_UseCase_YourStudentID.png
03_Flowchart_YourStudentID.drawio	03_Flowchart_YourStudentID.png

Saving and exporting — step by step

01 Save the diagram

While the diagram is open, use File > Save (or Ctrl+S). When saving to Device, confirm that a .drawio file is on your computer.

02 Export a picture

Choose File > Export as > PNG. Confirm the export options and click Export. Download or save the image to the same folder.

03 Check the files

Open the PNG to confirm the entire diagram is visible. Open the .drawio file through File > Open From > Device if you need to confirm it is editable.

04 Submit your individual folder

Follow your instructor's classroom submission method. Do not include another student's files. Do not submit only screenshots: the editable .drawio files are mandatory.

Assessment rubric — 10 marks

CRITERION	MARKS
Requirements mind map: classification, completeness and legibility	3
UML use-case diagram: actors, boundary, use cases and links	3
Process flowchart: sequence, decision and both paths	3
Correct filenames and all six deliverables	1

08 | Semester Project

TEAM ACTIVITY FOR THE SEMESTER

Form a team of three students

form a team with two other classmates. Choose ONE realistic software system that your team can study, design, plan, and improve over the semester. You do not need to develop the full application today.

Suggested project choices

OPTION	EXAMPLE FEATURES
Campus Lost-and-Found System	Report an item, search items, claim an item
Clinic Appointment System	View available times, request booking, cancel
Student Event Registration	Discover events, register, view attendance
Dormitory Maintenance Portal	Submit a repair request, assign it, track status
University Course Feedback	Complete survey, view aggregated results

Your team's first decision

01 Choose one problem

Pick a real and understandable problem relevant to university life. Keep the first version small enough to design within a semester.

02 Define the users

List at least two user roles, such as Student and Administrator, and explain their needs.

03 Agree on the proposed scope

Write 4–6 main features and at least two things the first version will NOT include.

04 Divide initial responsibilities

Assign a coordinator, a requirements lead, and a diagrams/documentation lead. These are starting roles; every member must contribute throughout the semester.

Team project proposal — complete after the lab

PROJECT TITLE	_____
MEMBER 1 / ID	_____
MEMBER 2 / ID	_____
MEMBER 3 / ID	_____
PROBLEM TO SOLVE	_____
MAIN USERS	_____
FOUR CORE FEATURES	_____

What happens next: In later labs, this same team project will be used for detailed requirements, scope and work breakdown structure, scheduling, risk management, project tracking, and presentation. Your instructor will specify the team proposal deadline separately.